Name: N. Tharun Sai

Reg No:192472382

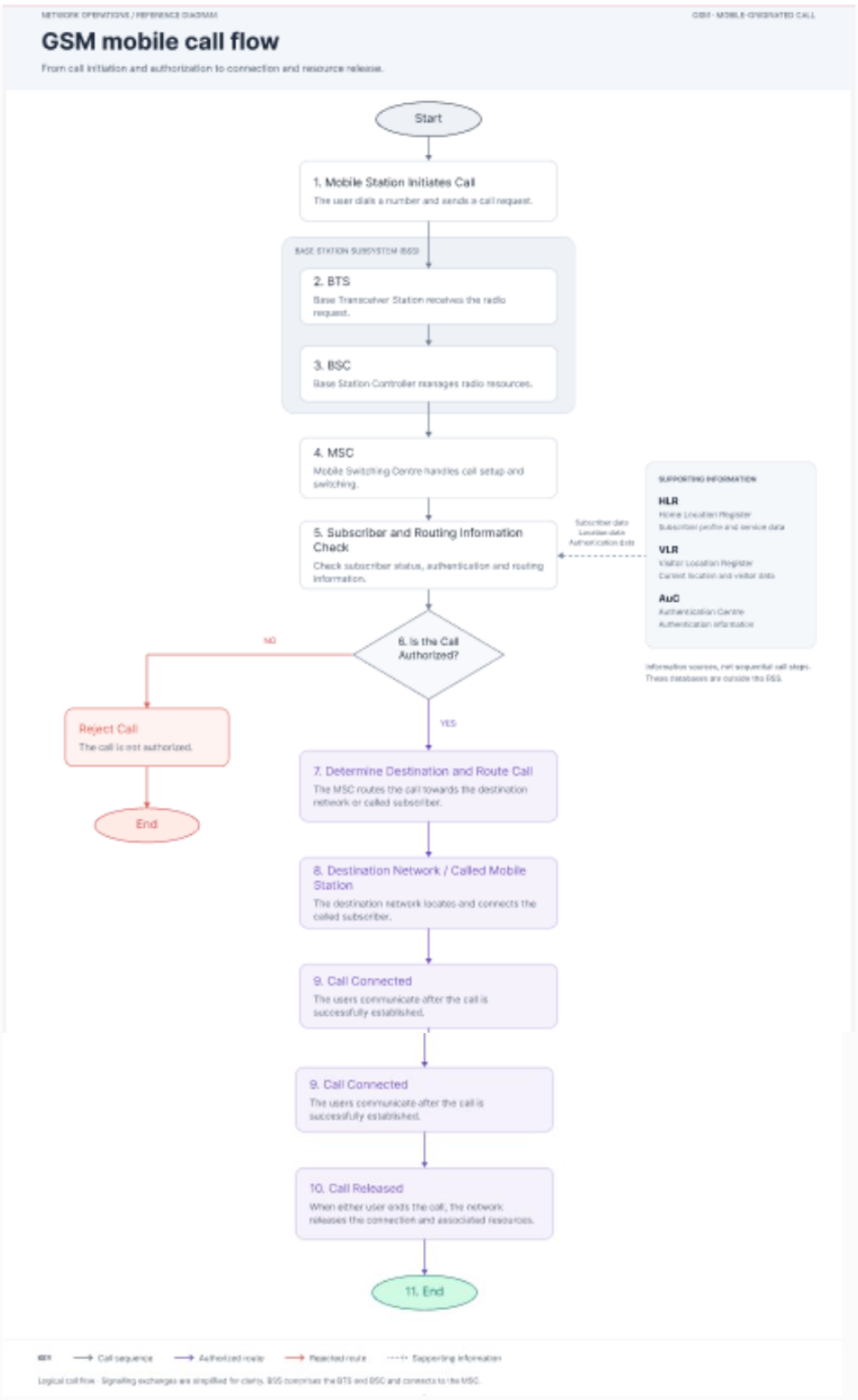
3. Develop a Flowchart in Figma Depicting the Operation of a Network Switching Subsystem (NSS)

AIM

To develop a flowchart using Figma that illustrates the operation of the Network Switching Subsystem (NSS) in a mobile cellular communication system

PROCEDURE

1. Open Figma and create a new design file.
2. Create a flowchart workspace using a suitable frame.
3. Add a **Start** symbol to indicate the beginning of the process.
4. Add a process block representing a mobile station initiating a call through the Base Transceiver Station (BTS) and Base Station Controller (BSC).
5. Add a process block for the Mobile Switching Centre (MSC), which processes the call request.
6. Include the Home Location Register (HLR) and Visitor Location Register (VLR) to retrieve subscriber and location information.
7. Add the Authentication Centre (AuC) to verify subscriber identity when required.
8. Add decision blocks to check subscriber authentication and destination availability.
9. Connect the blocks using arrows to represent call routing and decision paths.
10. Add a process block for establishing the connection between the calling and receiving mobile stations.
11. Include an **End** symbol to indicate call termination and release of network resources.
12. Review the flowchart and arrange the symbols neatly for clarity.


GSM mobile call flow
From call initiation and authorization to connection and resource release.
Start
1. Mobile Station Initiates Call
The user dials a number and sends a call request.
BASE STATION SUBSYSTEM (BSS)
2. BTS
Base Transceiver Station receives the radio request.
3. BSC
Base Station Controller manages radio resources.
4. MSC
Mobile Switching Centre handles call setup and switching.
5. Subscriber and Routing Information Check
Check subscriber status, authentication and routing information.
HLR
Home Location Register
VLR
Visitor Location Register
AuC
Authentication Centre
6. Is the Call Authorized?
NO
YES
Reject Call
The call is not authorized.
End
7. Determine Destination and Route Call
The MSC routes the call towards the destination network or called subscriber.
8. Destination Network / Called Mobile Station
The destination network locates and connects the called subscriber.
9. Call Connected
The users communicate after the call is successfully established.
9. Call Connected
The users communicate after the call is successfully established.
10. Call Released
When either user ends the call, the network releases the connection and associated resources.
11. End

RESULT

Thus, a flowchart representing the operation of the Network Switching Subsystem in a mobile cellular system was developed using Figma, illustrating subscriber authentication, location information retrieval, call routing, connection establishment, and call termination.